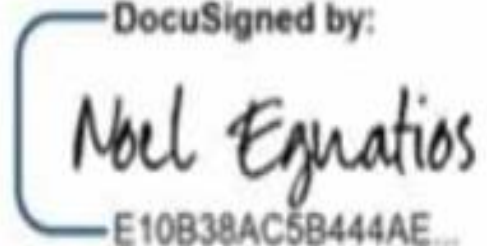


Unit 2

Examples of electronic signatures

 (Signature using a touch pad device)	DocuSigned by:  (Signature + Membership ID)
 (Signature using a mouse)	 (Signature (image))

Signature valid

Digitally signed by PlotSoft LLC
Date: 2013.05.06 21:12:36 +05:00
Reason: I'm the author
Location: USA



Valid signature

Signature valid

Dokument podpisany przez
Tomasz [redacted]
Data: 2025.12.30 11:44:09
CET



Digitally signed by fCoder SIA
Reason: I am the author
Location: Riga



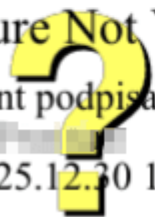
Signature invalid

Dokument podpisany przez
Tomasz [redacted]
Data: 2025.12.30 11:44:09
CET



Signature Not Verified

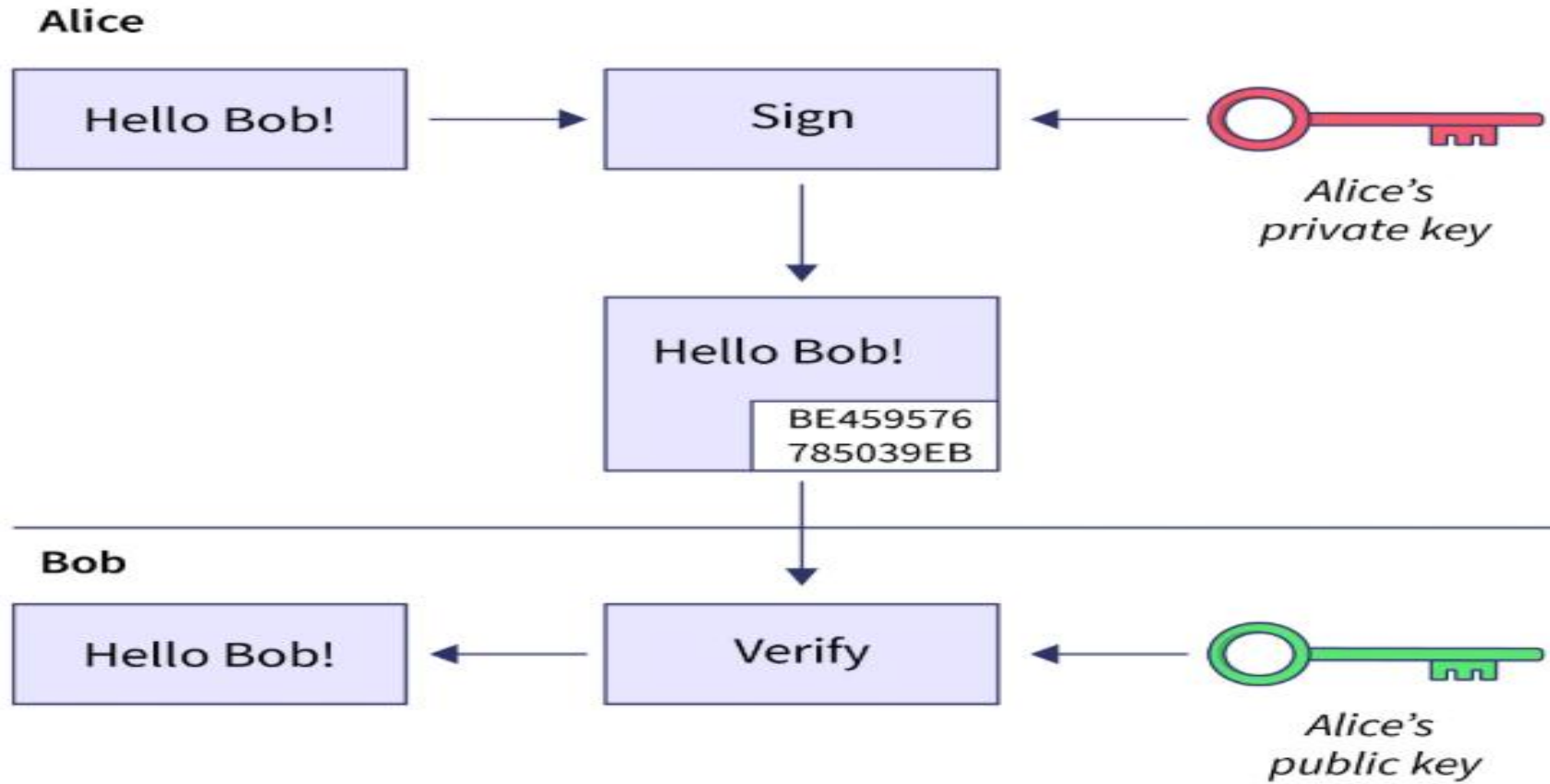
Dokument podpisany przez
Tomasz [redacted]
Data: 2025.12.30 11:44:09
CET

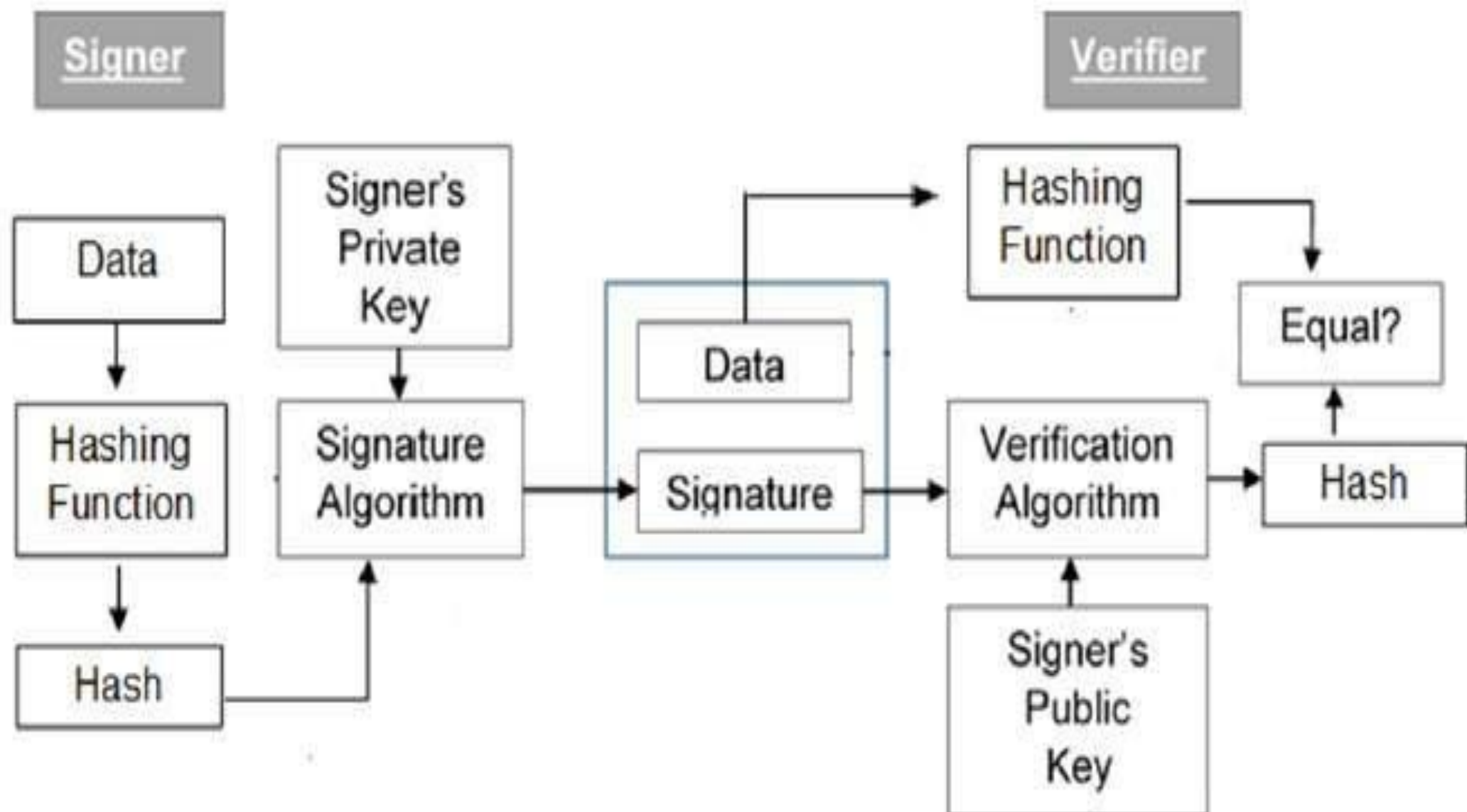


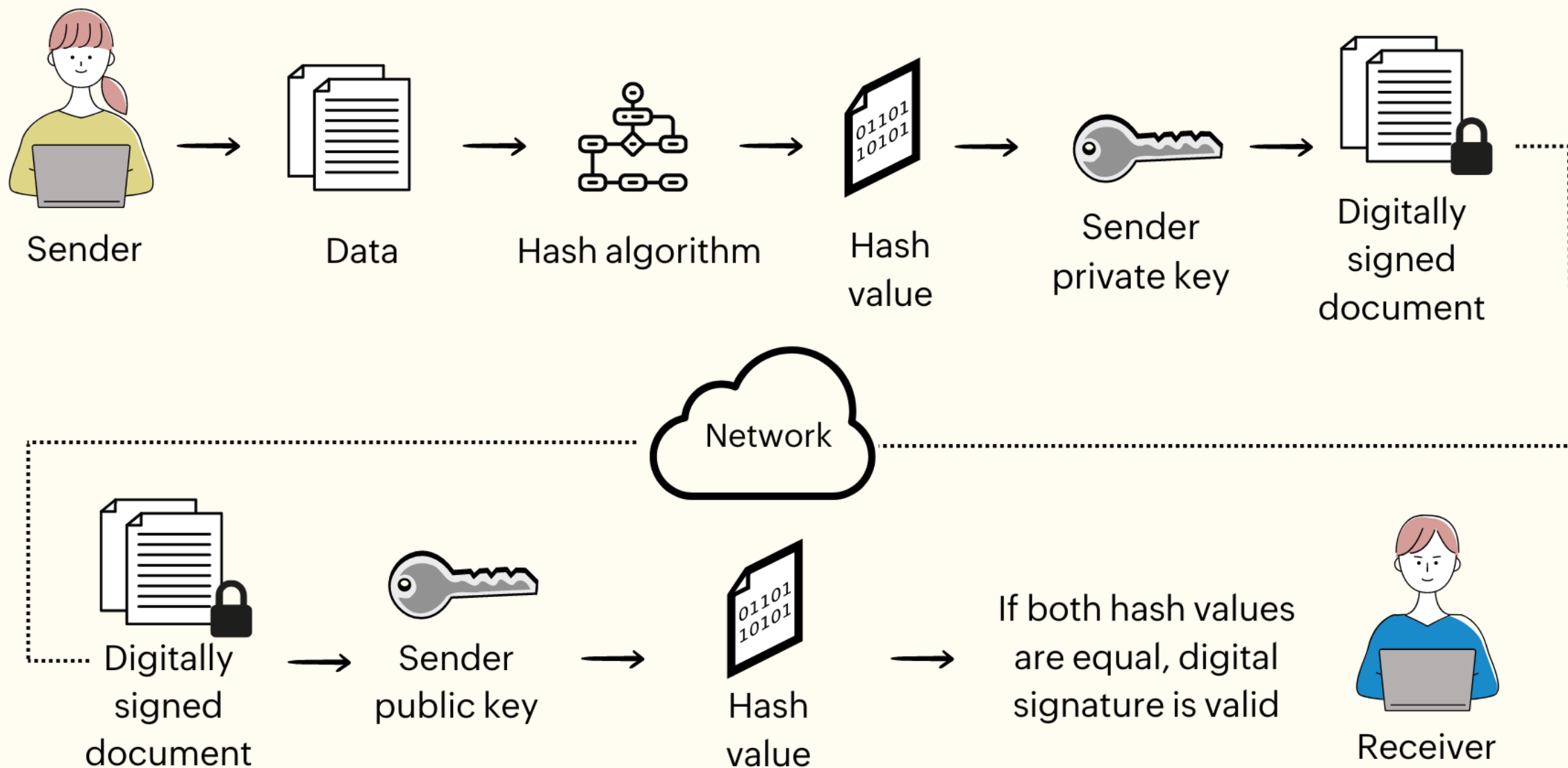
Validity unknown

Dokument podpisany przez
Tomasz [redacted]
Data: 2025.12.30 11:44:09
CET

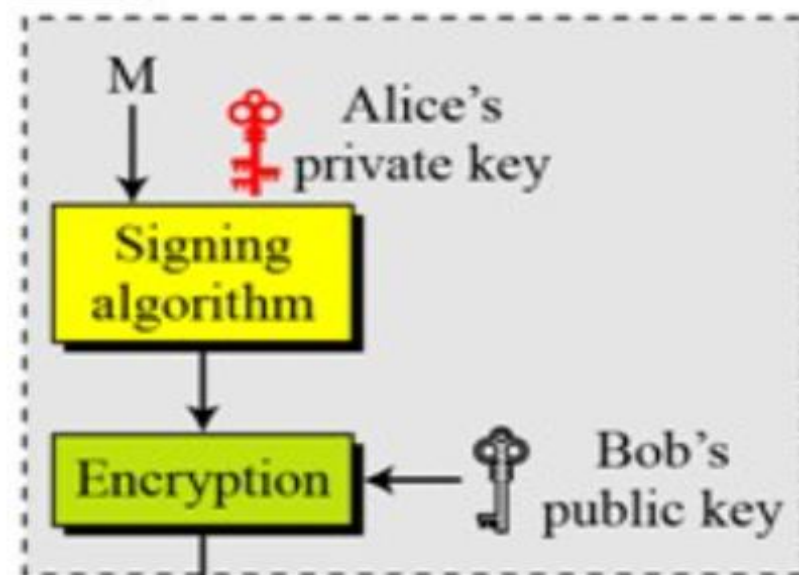






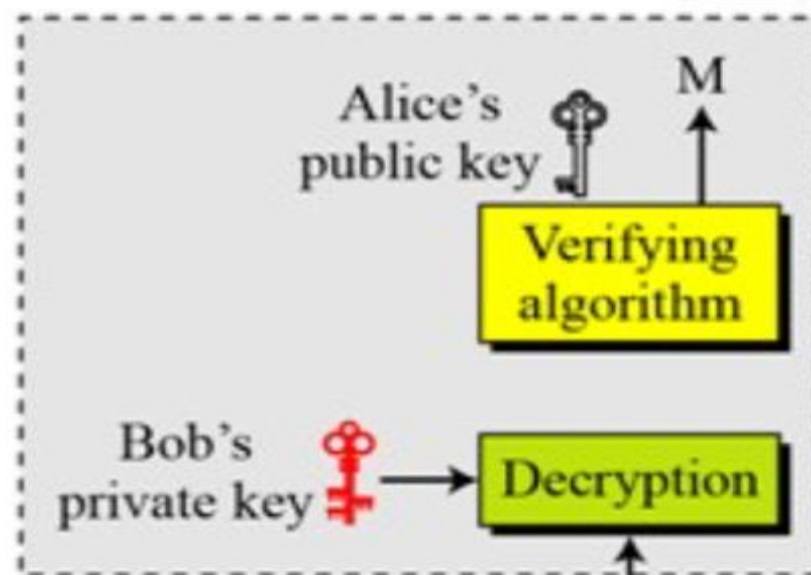


Alice



M: Message
S: Signature

Bob



Encrypted (M, S)



SSL/TLS



HTTPS



SSH



IPSec



S/MIME



Kerberos



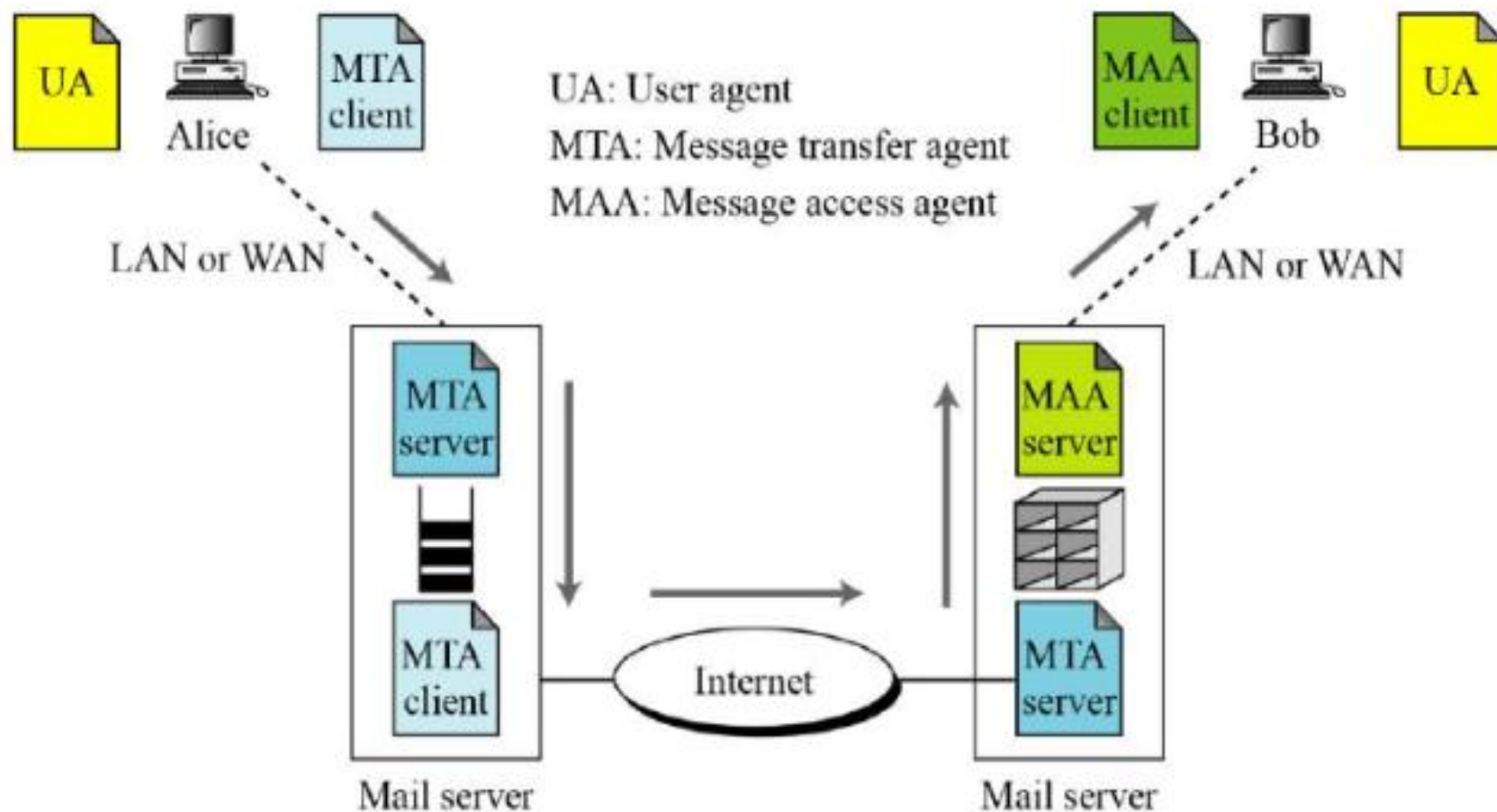
SSL VPN

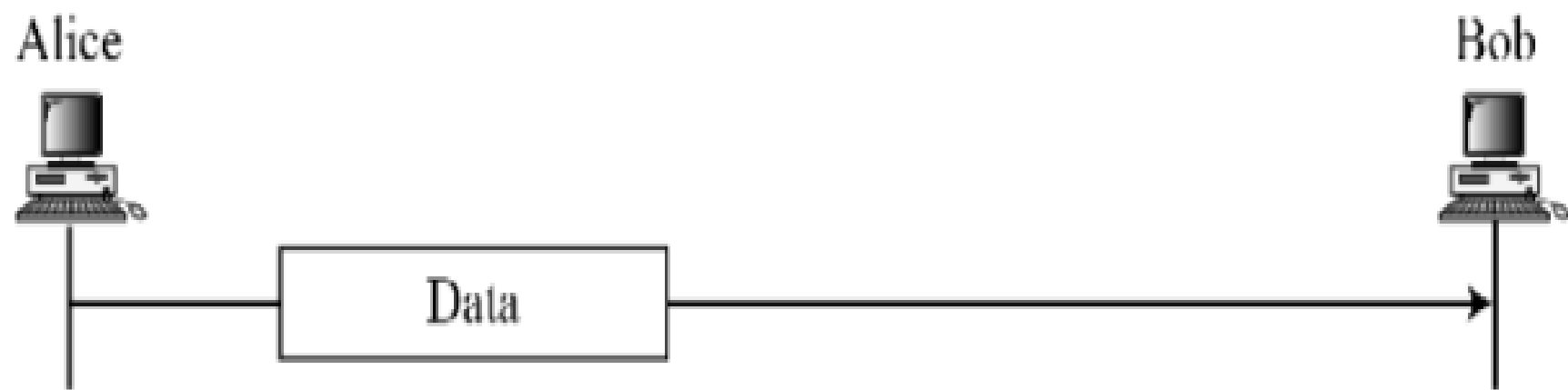


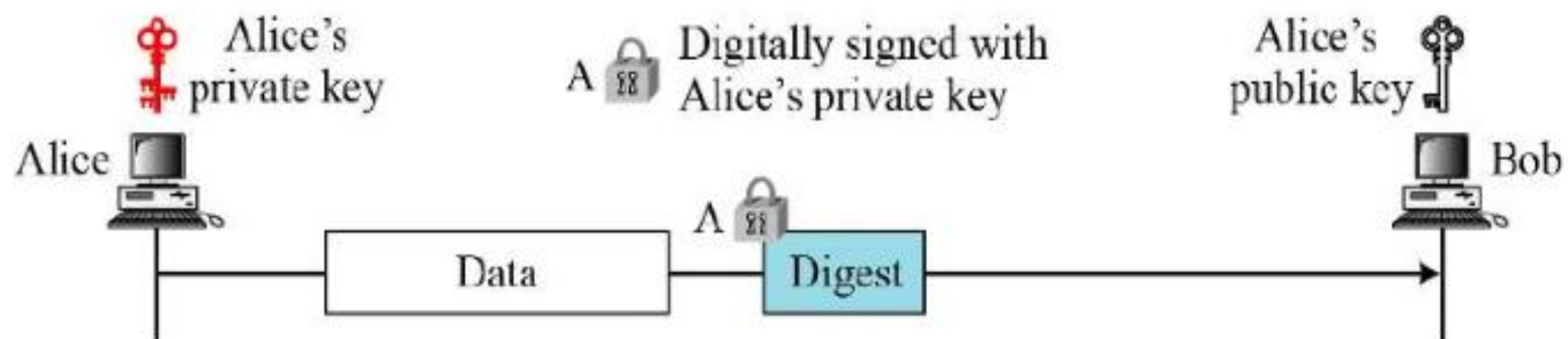
DNSSEC

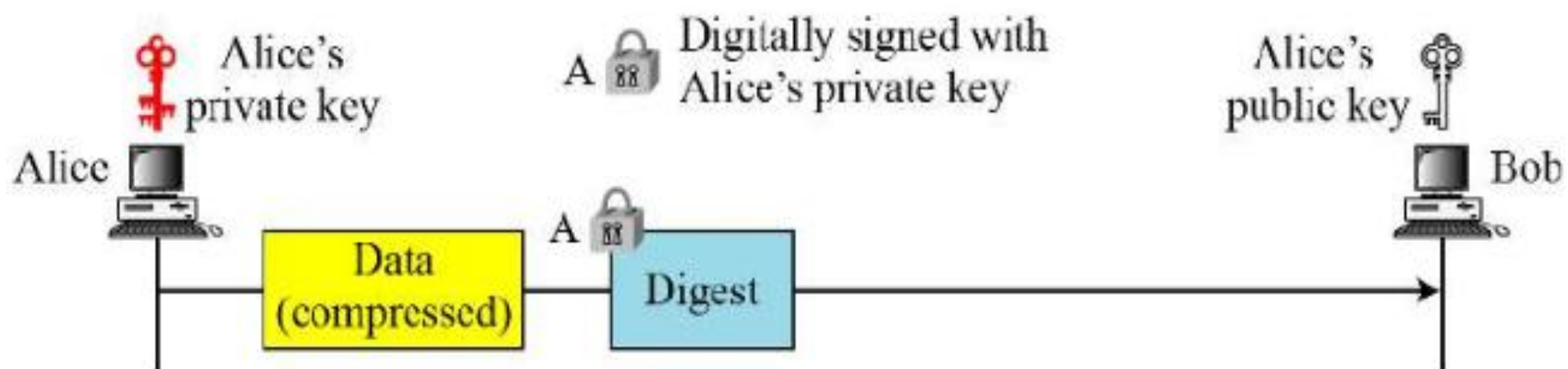
OSI MODEL LAYERS

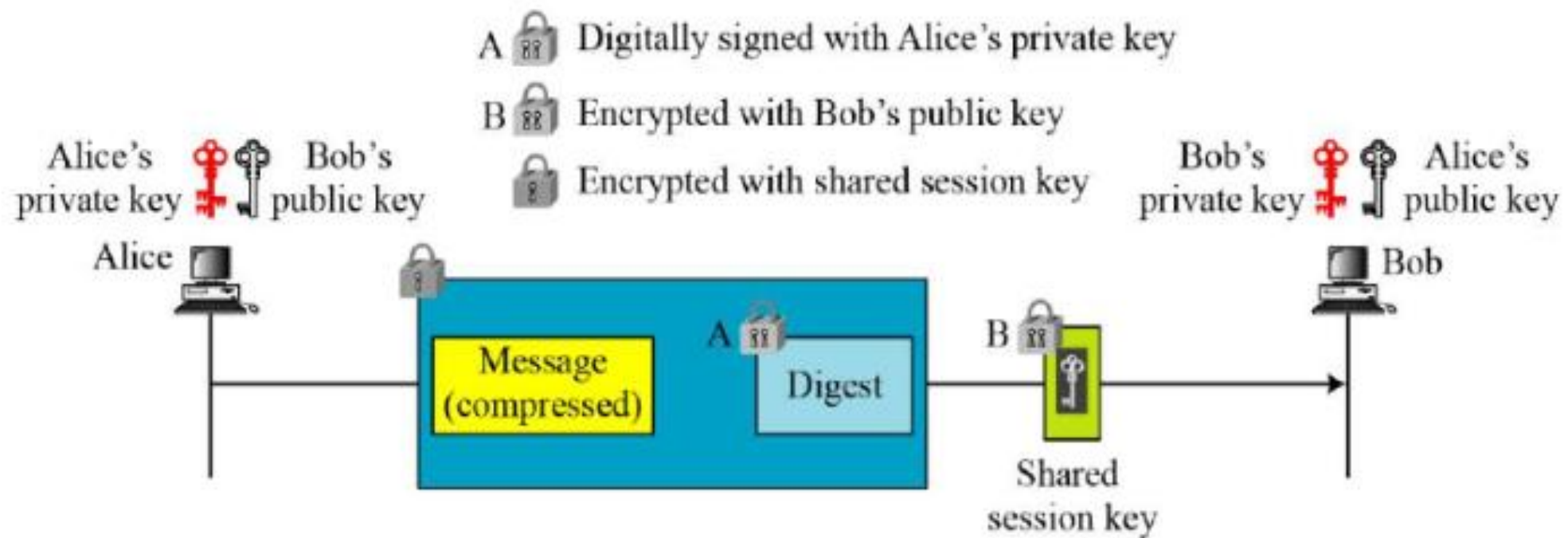
Application Layer	 Request Response HTTP, FTP, SMTP 
Presentation Layer	compression encoding encryption TLS, SSL
Session Layer	 Session Sockets 
Transport Layer	01011 101 Segmentation TCP, UDP Reassembly 01011 101
Network Layer	 Packets IP, ICMP, IPsec Packets assembly 
Data Link Layer	 Frames Ethernet, Wifi Intra-network communications 
Physical Layer	 Sending cable 00100111 Fiber Receiving cable 

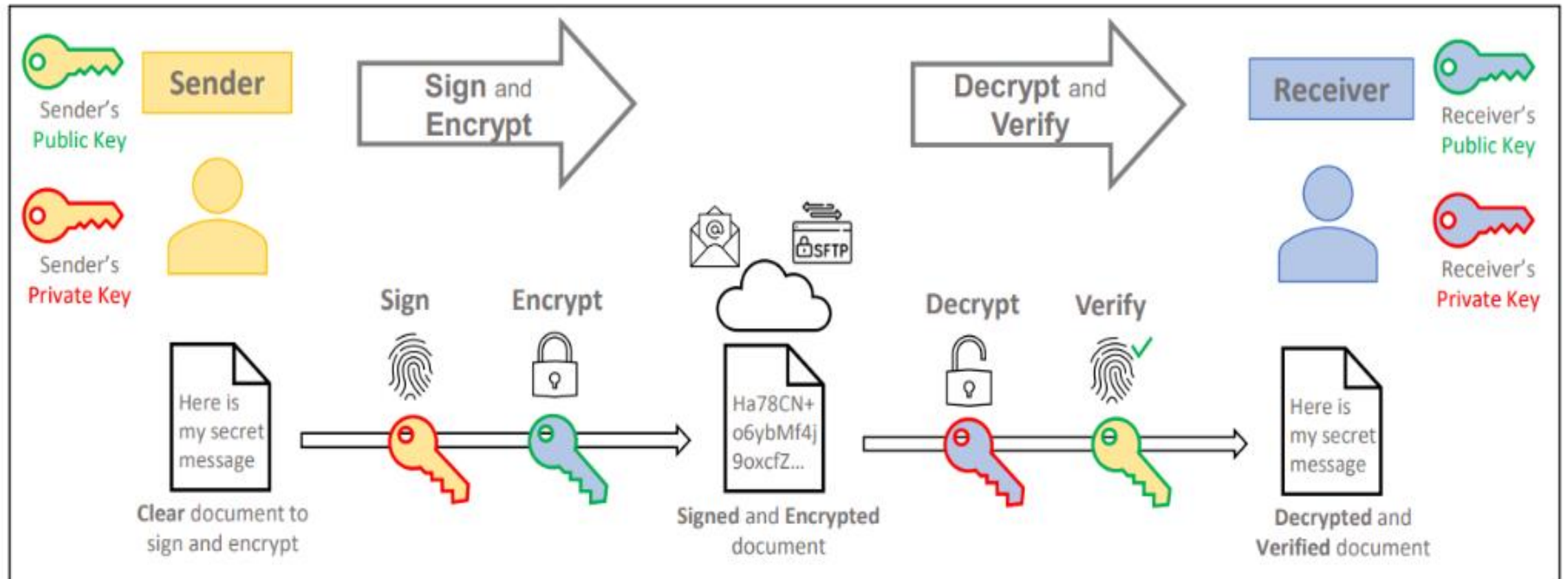










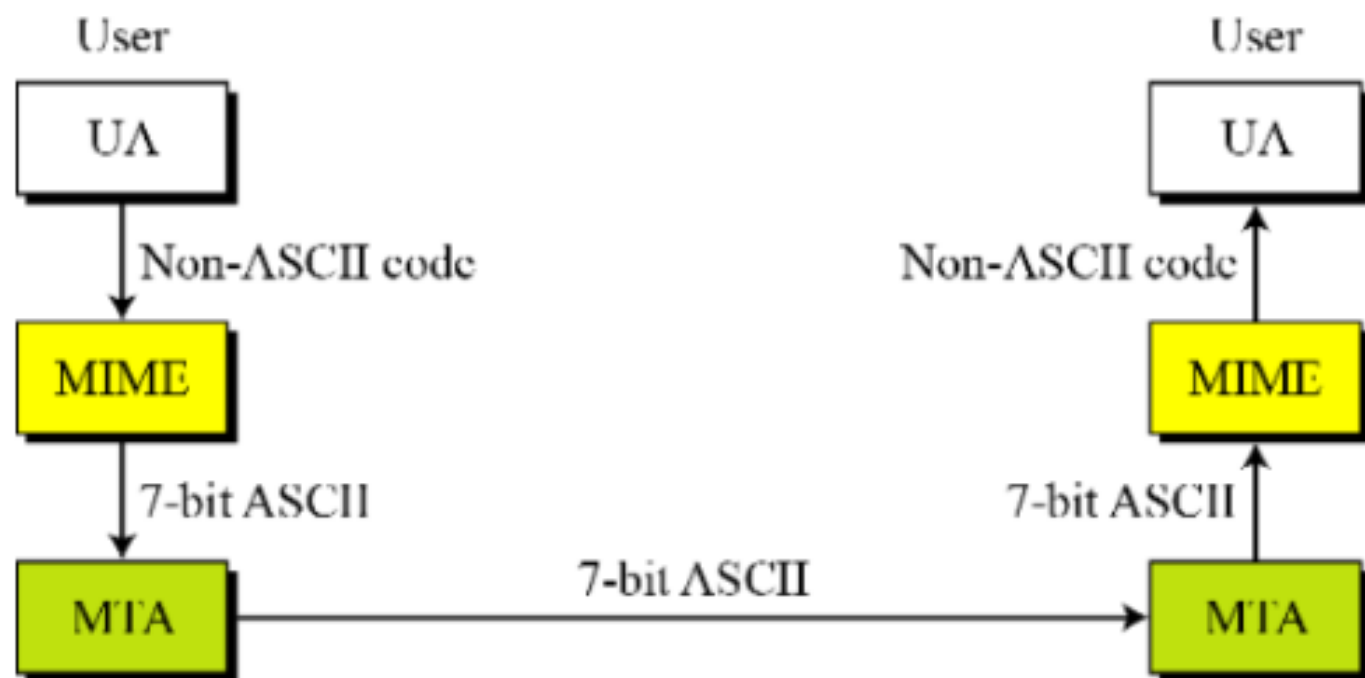


<i>ID</i>	<i>Description</i>
1	RSA (encryption or signing)
2	RSA (for encryption only)
3	RSA (for signing only)
16	ElGamal (encryption only)
17	DSS
18	Reserved for elliptic curve
19	Reserved for ECDSA
20	ElGamal (for encryption or signing)
21	Reserved for Diffie-Hellman
100–110	Private algorithms

<i>ID</i>	<i>Description</i>
0	No Encryption
1	IDEA
2	Triple DES
3	CAST-128
4	Blowfish
5	SAFER-SK128
6	Reserved for DES/SK
7	Reserved for AES-128
8	Reserved for AES-192
9	Reserved for AES-256
100–110	Private algorithms

<i>ID</i>	<i>Description</i>
1	MD5
2	SHA-1
3	RIPE-MD/160
4	Reserved for double-width SHA
5	MD2
6	TIGER/192
7	Reserved for HAVAL
100–110	Private algorithms

<i>ID</i>	<i>Description</i>
0	Uncompressed
1	ZIP
2	ZLIP
100–110	Private methods

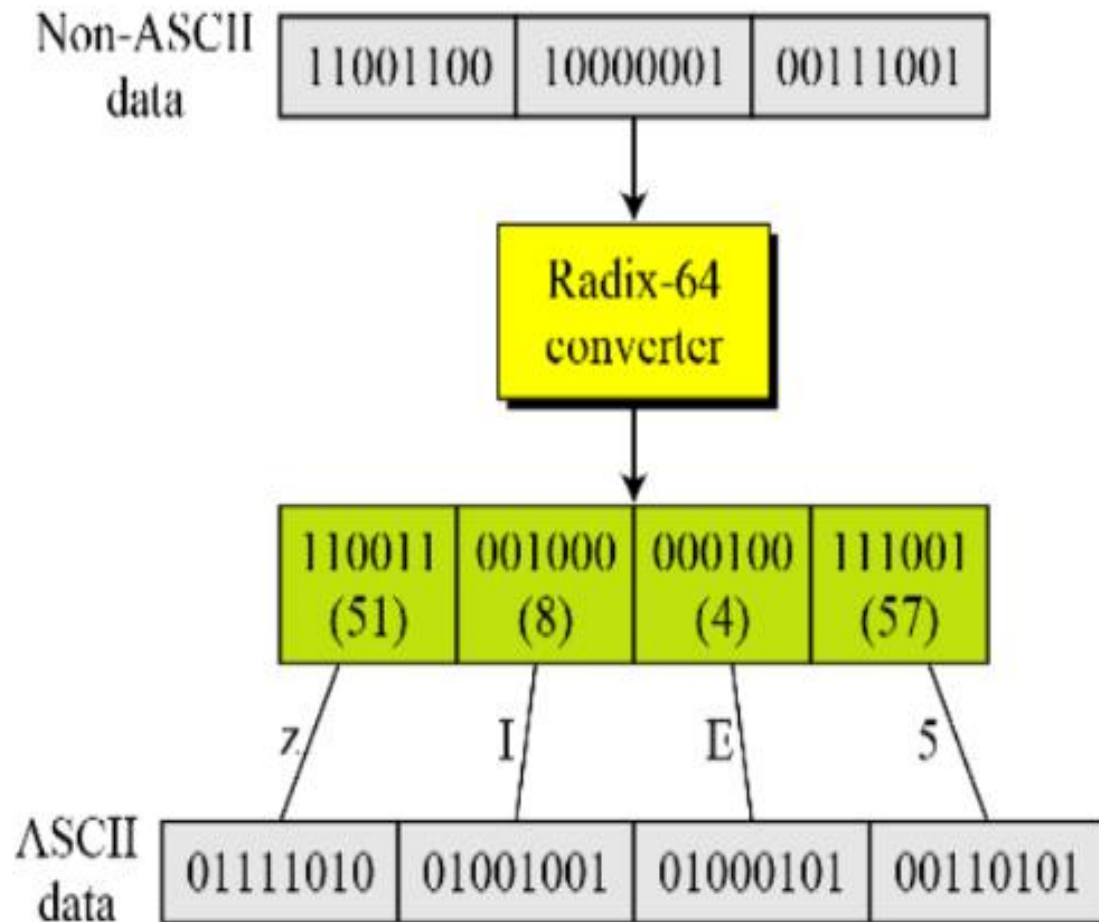


E-mail header
MIME-Version: 1.1 Content-Type: type/subtype Content-Transfer-Encoding: encoding type Content-Id: message id Content-Description: textual explanation of nontextual contents
E-mail body

MIME headers

<i>Type</i>	<i>Subtype</i>	<i>Description</i>
	Plain	Unformatted.
	HTML	HTML format.
Multipart	Mixed	Body contains ordered parts of different data types.
	Parallel	Same as above, but no order.
	Digest	Similar to Mixed, but the default is message/RFC822.
	Alternative	Parts are different versions of the same message.
Message	RFC822	Body is an encapsulated message.
	Partial	Body is a fragment of a bigger message.
	External-Body	Body is a reference to another message.
Image	JPEG	Image is in JPEG format.
	GIF	Image is in GIF format.
Video	MPEG	Video is in MPEG format.
Audio	Basic	Single channel encoding of voice at 8 KHz.
Application	PostScript	Adobe PostScript.
	Octet-stream	General binary data (eight-bit bytes).

<i>Type</i>	<i>Description</i>
7bit	NVT ASCII characters and short lines.
8bit	Non-ASCII characters and short lines.
Binary	Non-ASCII characters with unlimited-length lines.
Radix-64	6-bit blocks of data are encoded into 8-bit ASCII characters using Radix-64 conversion.
Quoted-printable	Non-ASCII characters are encoded as an equal sign followed by an ASCII code.



Value	Code	Value	Code	Value	Code	Value	Code	Value	Code	Value	Code
0	A	11	L	22	W	33	h	44	s	55	3
1	B	12	M	23	X	34	i	45	t	56	4
2	C	13	N	24	Y	35	j	46	u	57	5
3	D	14	O	25	Z	36	k	47	v	58	6
4	E	15	P	26	a	37	l	48	w	59	7
5	F	16	Q	27	b	38	m	49	x	60	8
6	G	17	R	28	c	39	n	50	y	61	9
7	H	18	S	29	d	40	o	51	z	62	+
8	I	19	T	30	e	41	p	52	0	63	/
9	J	20	U	31	f	42	q	53	1		
10	K	21	V	32	g	43	r	54	2		

00100110 &	01001100 l.	10011101 Non-ASCII	00111001 9	01001011 K	Mixed ASCII and non-ASCII data
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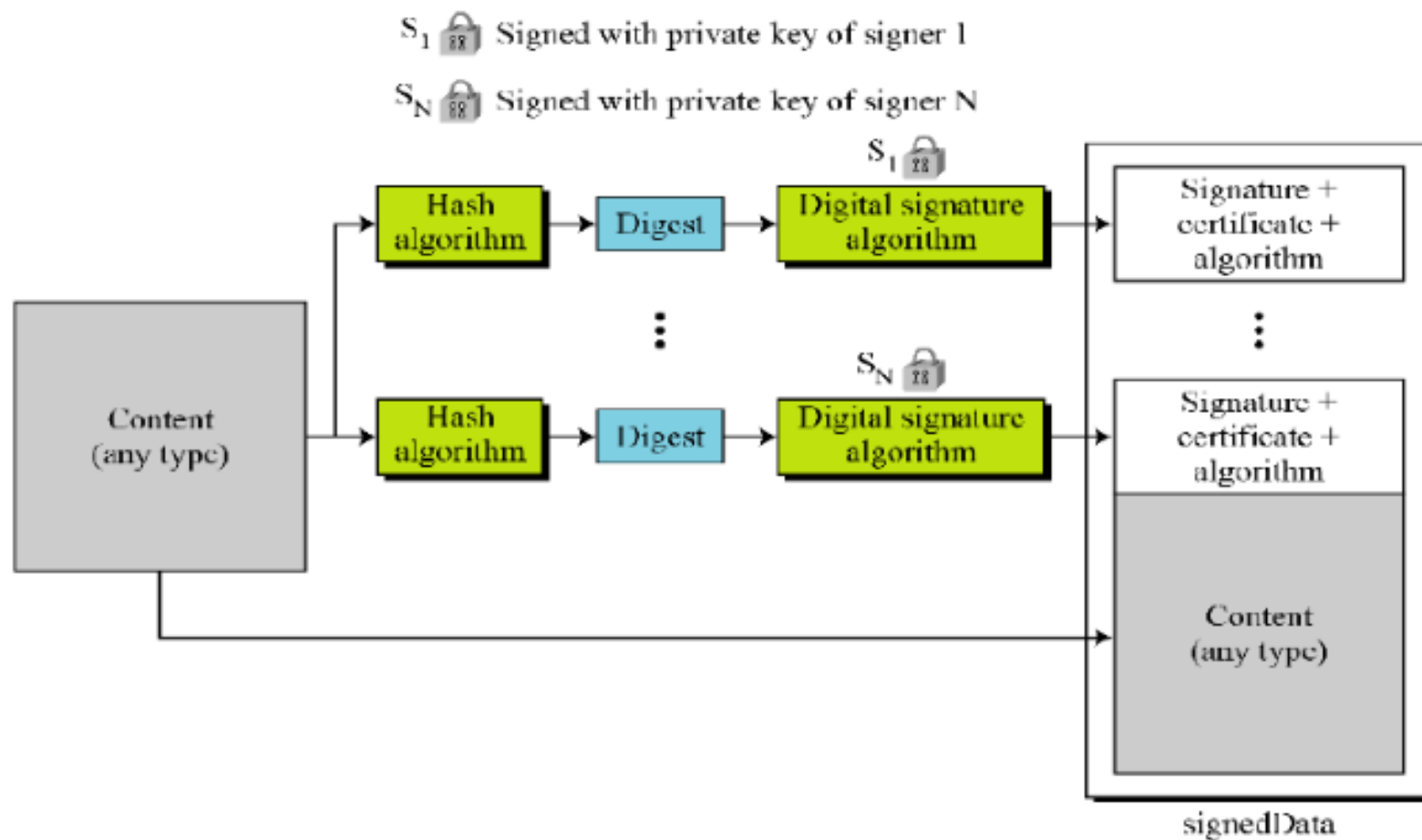


Quoted-
printable

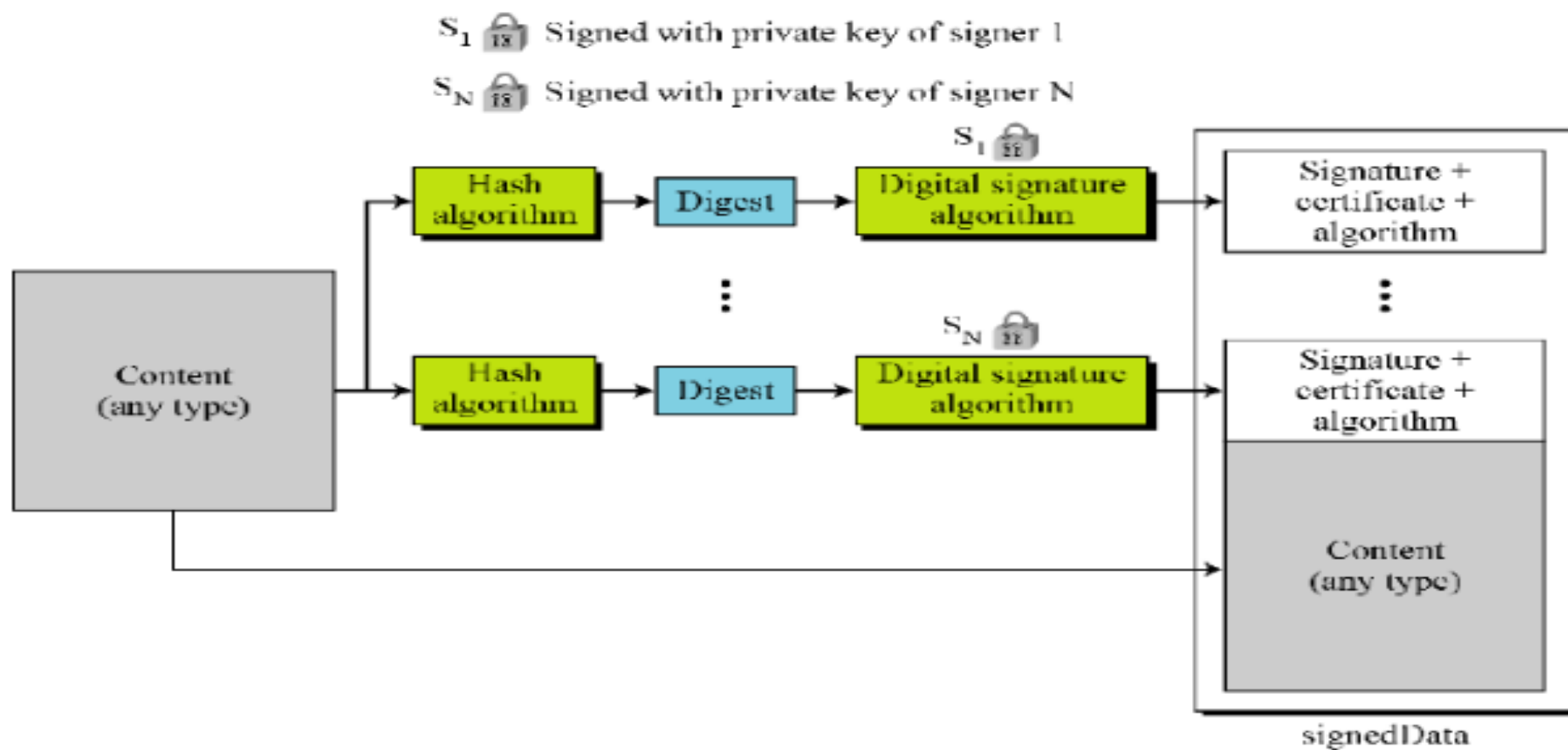


00100110 &	01001100 L	00111101 =	00111001 9	01000100 D	00111001 9	01001011 K	ASCII data
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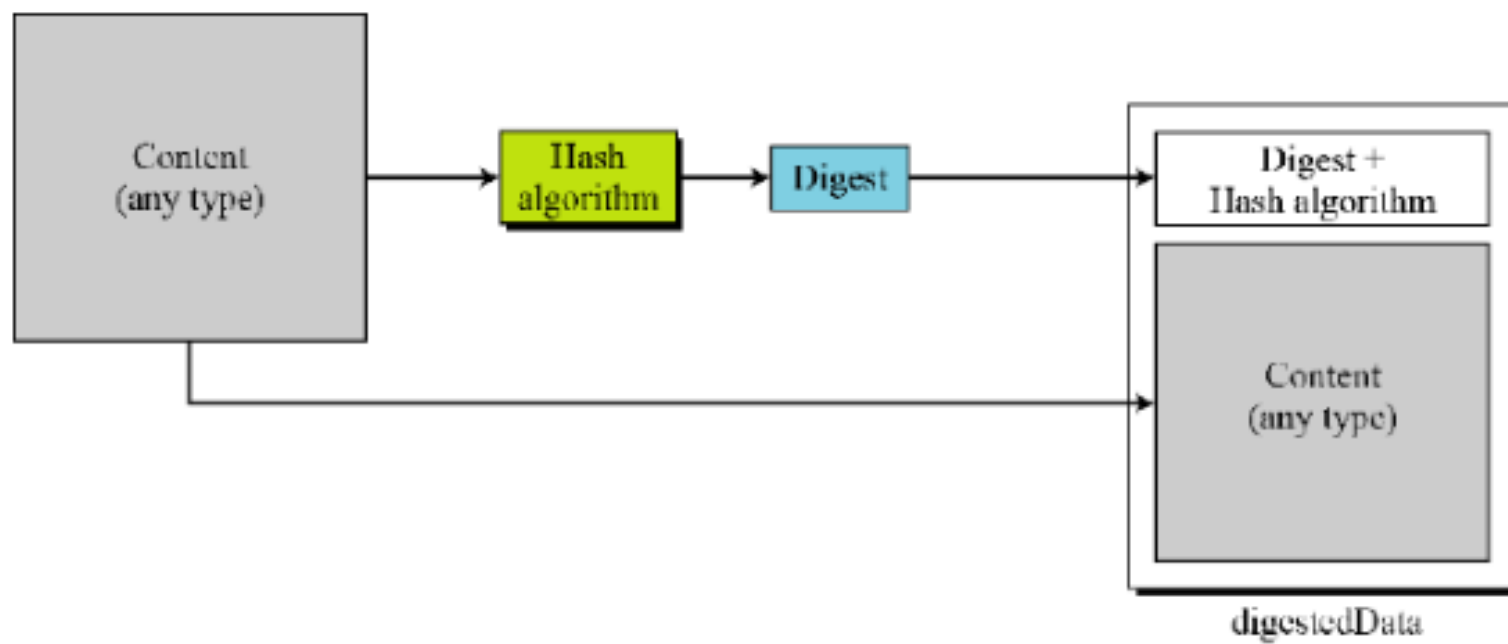
Signed-Data Content type



Enveloped-Data Content type



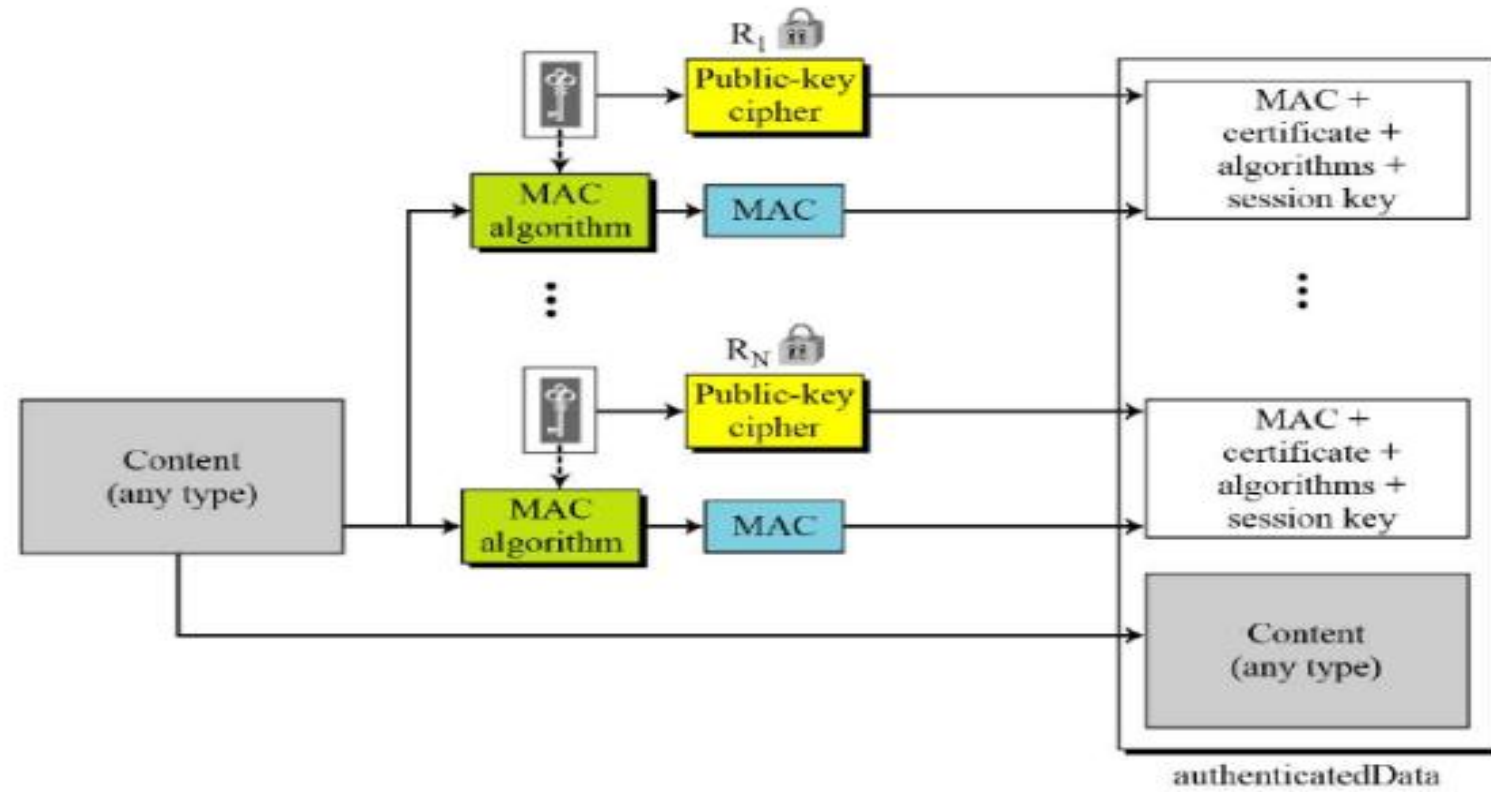
Digest-Data Content type



Authenticated data content type

R_1  Encrypted with public key of recipient 1

R_N  Encrypted with public key of recipient N



<i>Algorithm</i>	<i>Sender must support</i>	<i>Receiver must support</i>	<i>Sender should support</i>	<i>Receiver should support</i>
Content-encryption algorithm	Triple DES	Triple DES		1. AES 2. RC2/40
Session-key encryption algorithm	RSA	RSA	Diffie-Hellman	Diffie-Hellman
Hash algorithm	SHA-1	SHA-1		MD5
Digest-encryption algorithm	DSS	DSS	RSA	RSA
Message-authentication algorithm		HMAC with SHA-1		

Content-Type: application/pkcs7-mime; mime-type=enveloped-data

Content-Transfer-Encoding: Radix-64

Content-Description: attachment

name="report.txt";

cb32ut67f4bhijHU21oi87eryb0287hmnklsgFDoY8bc659GhIGfH6543mhjkdsaH23YjBnmN
ybmlkzjhgfdyhGe23Kjk34XiuD678Es16se09jy76jHuytTMDcbnmlkjgfFdiuyu678543m0n3h
G34un12P2454Hoi87e2ryb0H2MjN6KuyrlsgFDoY897fk923jljk1301XiuD6gh78EsUyT23y

